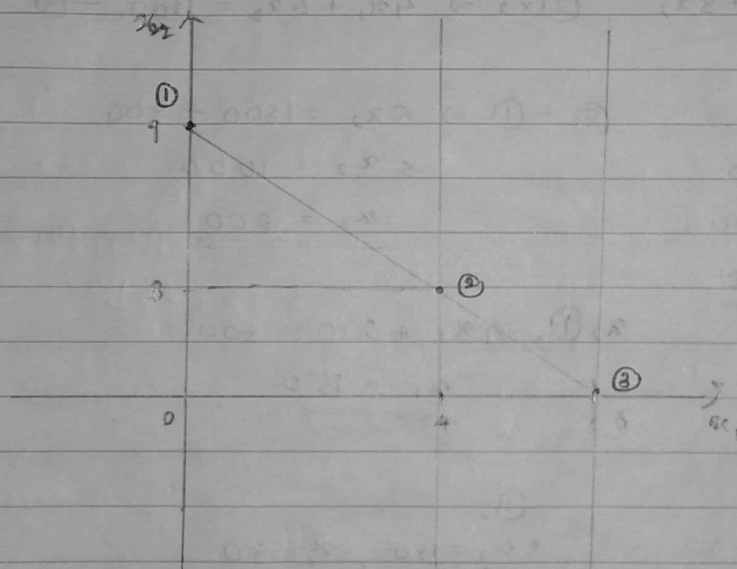


IM / 2023/091

Mathurathree M. C. S

Tutorial 1.1(01) Maximum : $Z = 3x_1 + 5x_2$ Subject to, $x_1 \leq 4$ $x_1 \geq 0$ $2x_1 \leq 12$ $x_2 \geq 0$ $3x_1 + 2x_2 \leq 18$ • $x_1 = 0, x_2 = 9$ • $x_1 = 4, x_2 = 3$ $x_2 = 0, x_1 = 6$ • $x_1 = 6, x_2 = 0$ • ① $\rightarrow Z = 3x_1 + 5x_2$ ② $\rightarrow Z = 3 \times 4 + 5 \times 3$ $Z = 0 + 5 \times 9$ $Z = 37$ $Z = 45$ ③ $\rightarrow Z = 3 \times 0 + 5 \times 6$ $Z = 30$ Maximum $Z = 45$ optimal $x_1 = 0$ optimal $x_2 = 9$

(02) Maximum : $Z = 6x_1 + 5x_2$

Subject to :

$$4x_1 + x_2 \leq 800 \quad \text{--- (1)}$$

$$2x_1 + 3x_2 \leq 900 \quad \text{--- (2)}$$

$$x_1 \leq 180$$

$$x_1 \geq 0, x_2 \geq 0$$

① (1), $x_1 = 0, x_2 = 800$

$x_2 = 0, x_1 = 200$

②, $x_1 = 0, x_2 = 300$

$x_2 = 0, x_1 = 450$

① = ②,

$$4x_1 + x_2 = 2x_1 + 3x_2$$

② $\times 2 \Rightarrow 4x_1 + 6x_2 = 1800$ --- (3)

$$2x_1 = 2x_2$$

$$x_1 = x_2$$

③ - ① $\Rightarrow 5x_2 = 1800 - 800$

$$5x_2 = 1000$$

$$x_2 = 200$$

①, $4x_1 + x_2 = 800$

$$5x_1 = 800$$

$$x_1 = 160$$

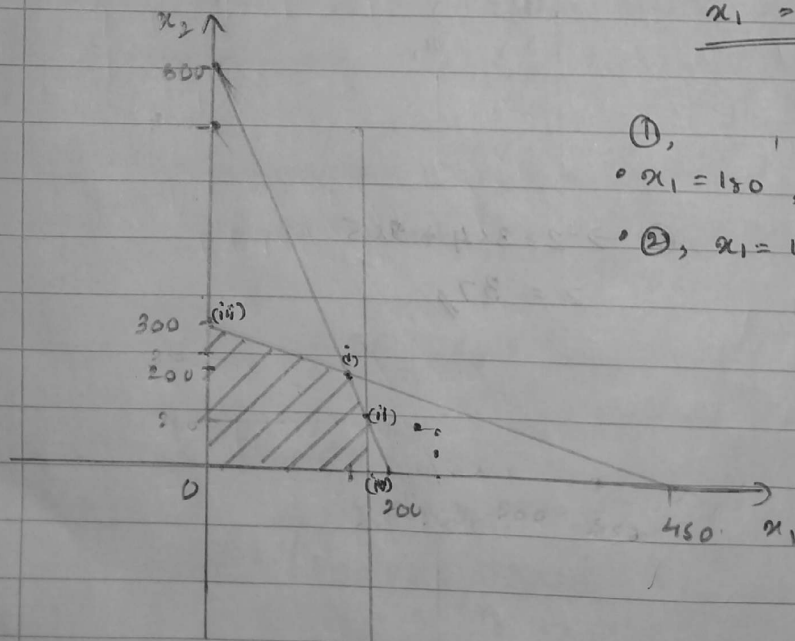
x_2 ①, $4x_1 + 200 = 800$

$$x_1 = 150$$

①,

$x_1 = 180, x_2 = 80$

②, $x_1 = 180, x_2 = 180$ --- (1)



Maximum $Z = 6 \times 150 +$

(i) $x_1 = 150, x_2 = 200$ (ii) $x_1 = 180, x_2 = 80$

$$z_1 = 6 \times 150 + 5 \times 200$$

$$z_2 = 6 \times 180 + 5 \times 80$$

$$z_1 = 900 + 1000$$

$$z_2 = 1080 + 400$$

$$z_1 = 1900$$

$$z_2 = 1480$$

(iii) $x_1 = 300, x_2 = 0$

(iv) $x_1 = 180, x_2 = 0$

$$z_3 = 6 \times 0 + 5 \times 300$$

$$z_4 = 6 \times 180 + 5 \times 0$$

$$z_3 = 1500$$

$$z_4 = 1080$$

Maximum z is 1900.

Optimal $x_1 = 150$

Optimal $x_2 = 200$

(03)	Basic Bq. x_B		Basic x_P	
sells	\$300	\$480	assembling	5 hours
cost of component	150	225	testing	1 hour

• assembling cost - 11 * cost for components =

testing cost - 15

• x_B - Number of Basics

• Basic requires ~~5h~~

x_{xP} - Number of x_P s

• Cost for components = $150x_B + 225x_{xP}$

• Cost for assemble and test = $[(5 \times 11)x_B + 15x_B] + [(6 \times 11)x_{xP} + (2 \times 15)x_{xP}]$

$$= 5570x_B + 96x_{xP}$$

• Objective function.

$$\text{PI Profit} = [300x_B + 480x_{xP}] - [70x_B + 96x_{xP}] + 150x_B + 225x_{xP}$$

$$\text{Profit} = 80x_B + 129x_{xP}$$

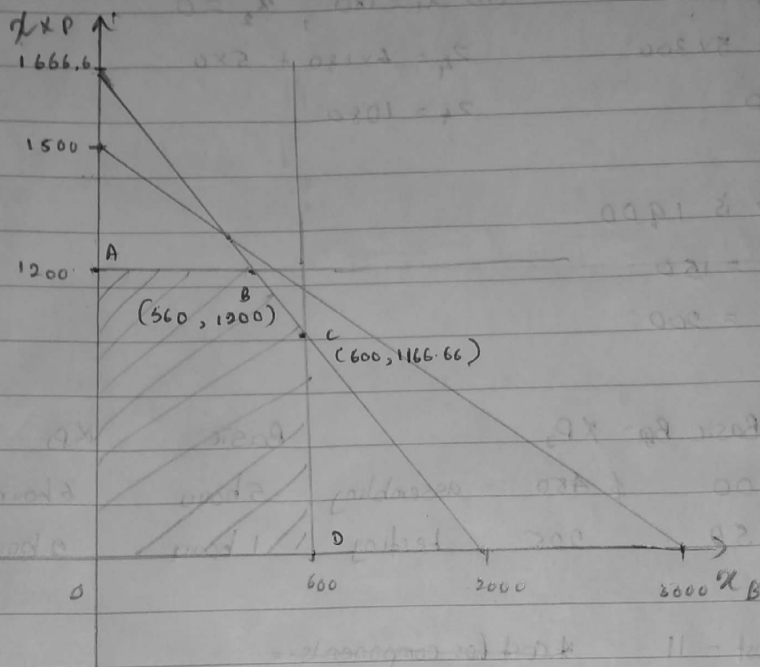
Constraints,

$$\bullet x_B \leq 600, x_{xp} \leq 1200$$

$$\bullet 5x_B + 6x_{xp} \leq 10000$$

$$\bullet x_B + 2x_{xp} \leq 3000$$

$$\bullet x_B \geq 0, x_{xp} \geq 0$$



$$\bullet 5x_B + 6x_{xp} = 10000$$

$$\bullet 5x_B + 6x_{xp} = 10000$$

$$5x_B + 6 \times 600 = 10000$$

$$x_{xp} = 1200$$

$$5x_B =$$

$$5x_B = 1000 - 6 \times 1200$$

$$x_B = 600; x_{xp} = 1166.66$$

$$x_B = 560$$

$$A \rightarrow x_B = 0, x_{xp} = 1200$$

$$Z_A = 154800$$

$$B \rightarrow x_B = 560, x_{xp} = 1200$$

$$Z_B = 199600$$

$$C \rightarrow x_B = 600, x_{xp} = 1166.66$$

$$Z_C = 198500$$

$$D \rightarrow x_B = 600, x_{xp} = 0$$

$$Z_D = 48000$$

$$\bullet \text{Maximum Profit} = 199600$$

$$\text{Optimal } x_B = 560$$

$$\text{Optimal } x_{xp} = 1200$$